

1. Arithmetic Operators :

```
a <- 10
```

```
b <- 3
```

```
a + b
```

```
a - b
```

```
a * b
```

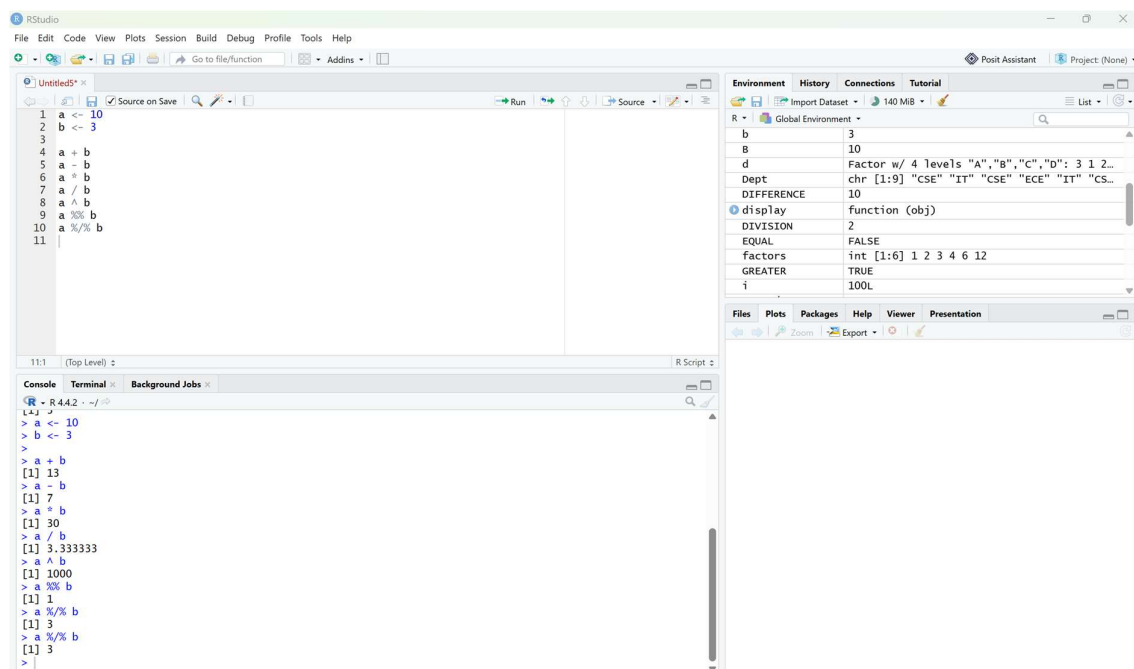
```
a / b
```

```
a ^ b
```

```
a %% b
```

```
a %/% b
```

Output :



The screenshot shows the RStudio interface. The script editor on the left contains the following code:

```
1 a <- 10
2 b <- 3
3
4 a + b
5 a - b
6 a * b
7 a / b
8 a ^ b
9 a %% b
10 a %/% b
11
```

The console on the right shows the output of these operations:

```
R - R 4.4.2 - ~/R
> a <- 10
> b <- 3
>
> a + b
[1] 13
> a - b
[1] 7
> a * b
[1] 30
> a / b
[1] 3.333333
> a ^ b
[1] 1000
> a %% b
[1] 1
> a %/% b
[1] 3
> a %/% b
[1] 3
>
```

The Environment pane on the right shows the current state of the workspace:

Object	Value
b	3
B	10
d	Factor w/ 4 levels "A","B","C","D": 3 1 2...
Dept	chr [1:9] "CSE" "IT" "CSE" "ECE" "IT" "CS..."
DIFFERENCE	10
display	function (obj)
DIVISION	2
EQUAL	FALSE
factors	int [1:6] 1 2 3 4 6 12
GREATER	TRUE
i	100L

2. Assignment Operators

```
a <- 10
```

```
b = 20
```

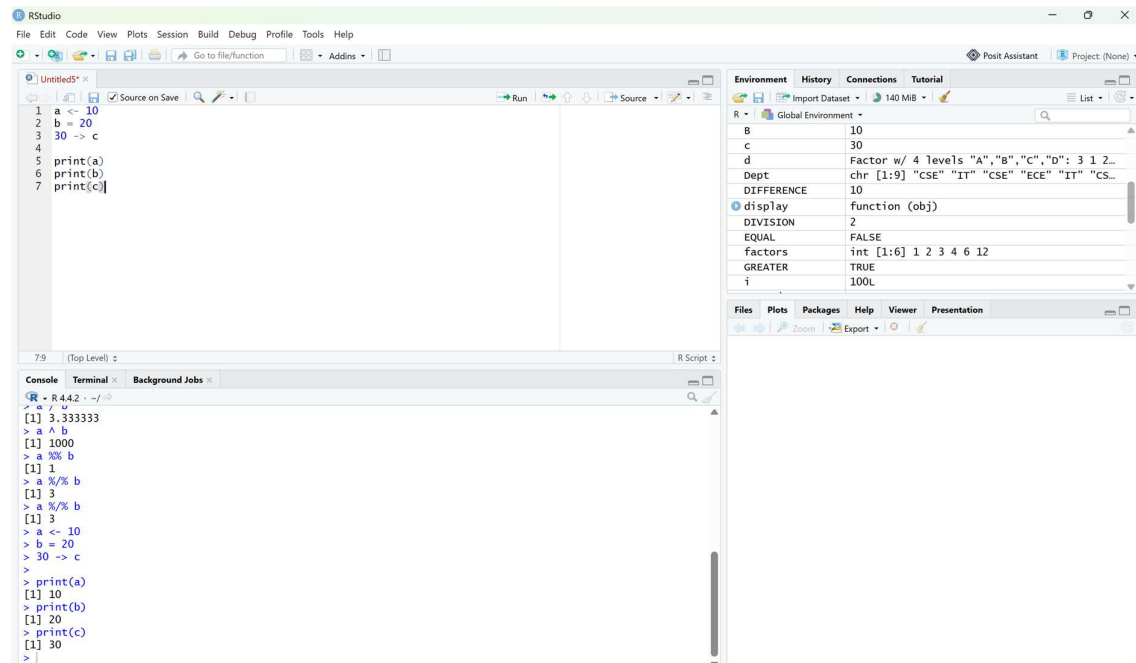
```
30 -> c
```

```
print(a)
```

```
print(b)
```

```
print(c)
```

Output :



The screenshot shows the RStudio interface. The source editor on the left contains the following R code:

```
1 a <- 10
2 b = 20
3 30 -> c
4
5 print(a)
6 print(b)
7 print(c)
```

The console on the bottom left shows the output of the code:

```
> R 4.4.2 ~ /
> a
[1] 5.333333
> a * b
[1] 1000
> a %/% b
[1] 1
> a %/% b
[1] 3
> a %/% b
[1] 3
> a <- 10
> b = 20
> 30 -> c
>
> print(a)
[1] 10
> print(b)
[1] 20
> print(c)
[1] 30
>
```

The Environment pane on the right shows the current state of the workspace:

Object	Value
B	10
c	30
d	Factor w/ 4 levels "A","B","C","D": 3 1 2...
Dept	chr [1:9] "CSE" "IT" "CSE" "ECE" "IT" "CS..."
DIFFERENCE	10
display	function (obj)
DIVISION	2
EQUAL	FALSE
factors	int [1:6] 1 2 3 4 6 12
GREATER	TRUE
i	100L

3. Comparison Operators

```
a <- 10
```

```
b <- 5
```

```
a == b
```

```
a != b
```

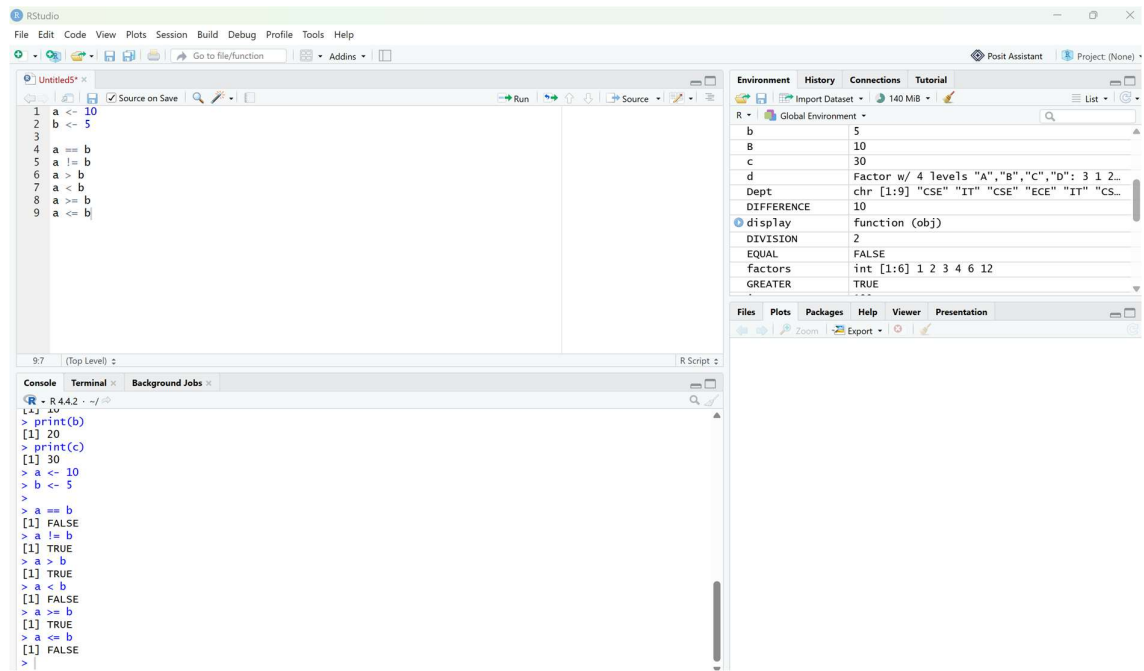
```
a > b
```

```
a < b
```

```
a >= b
```

```
a <= b
```

Output :



4. Logical Operators

a <- TRUE

b <- FALSE

a & b

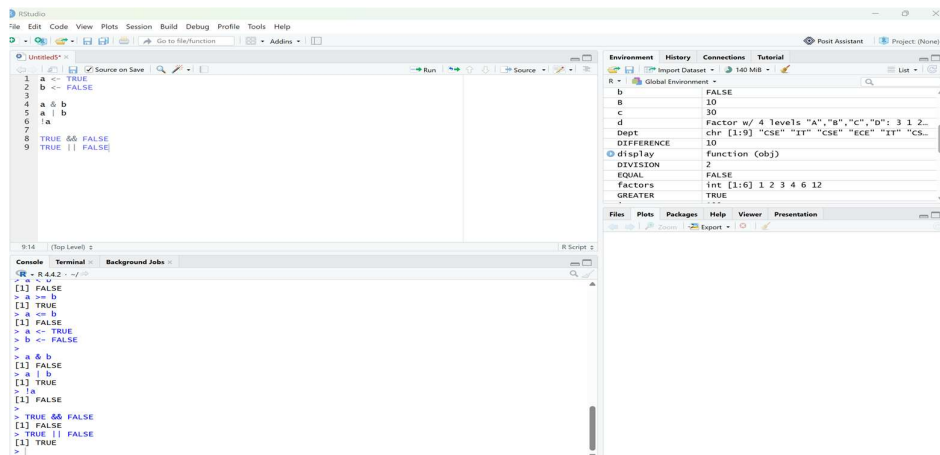
a | b

!a

TRUE && FALSE

TRUE || FALSE

Output :



5. Miscellaneous Operators:

```
x <- 1:5
```

```
3 %in% x
```

```
1:5
```

```
5:1
```

```
x <- 1:5
```

```
y <- 6:10
```

```
x %in% y
```

```
x %%% y
```

Output :

The screenshot shows the RStudio interface with the following content:

Source Editor:

```
1 x <- 1:5
2
3 3 %in% x
4
5 1:5
6
7 5:1
8
9 x <- 1:5
10 y <- 6:10
11
12 x %in% y
13
14 x %%% y
```

Console:

```
>
> 3 %in% x
[1] TRUE
>
> 1:5
[1] 1 2 3 4 5
>
> 5:1
[1] 5 4 3 2 1
>
> x <- 1:5
> y <- 6:10
>
> x %in% y
[1] FALSE FALSE FALSE FALSE FALSE
>
> x %%% y
[1,] 130
> |
```

Environment:

min_mark	70
n	12L
Name	"Vahidh"
overall_average	81.7666666666667
PRODUCT	200
RegNo	num [1:9] 101 102 103 104 105 106 107 108...
student_average	Named num [1:10] 84.3 70 87 77.7 92.3 ...
subject	"Chemistry"
SUM	30
x	int [1:5] 1 2 3 4 5
y	int [1:5] 6 7 8 9 10